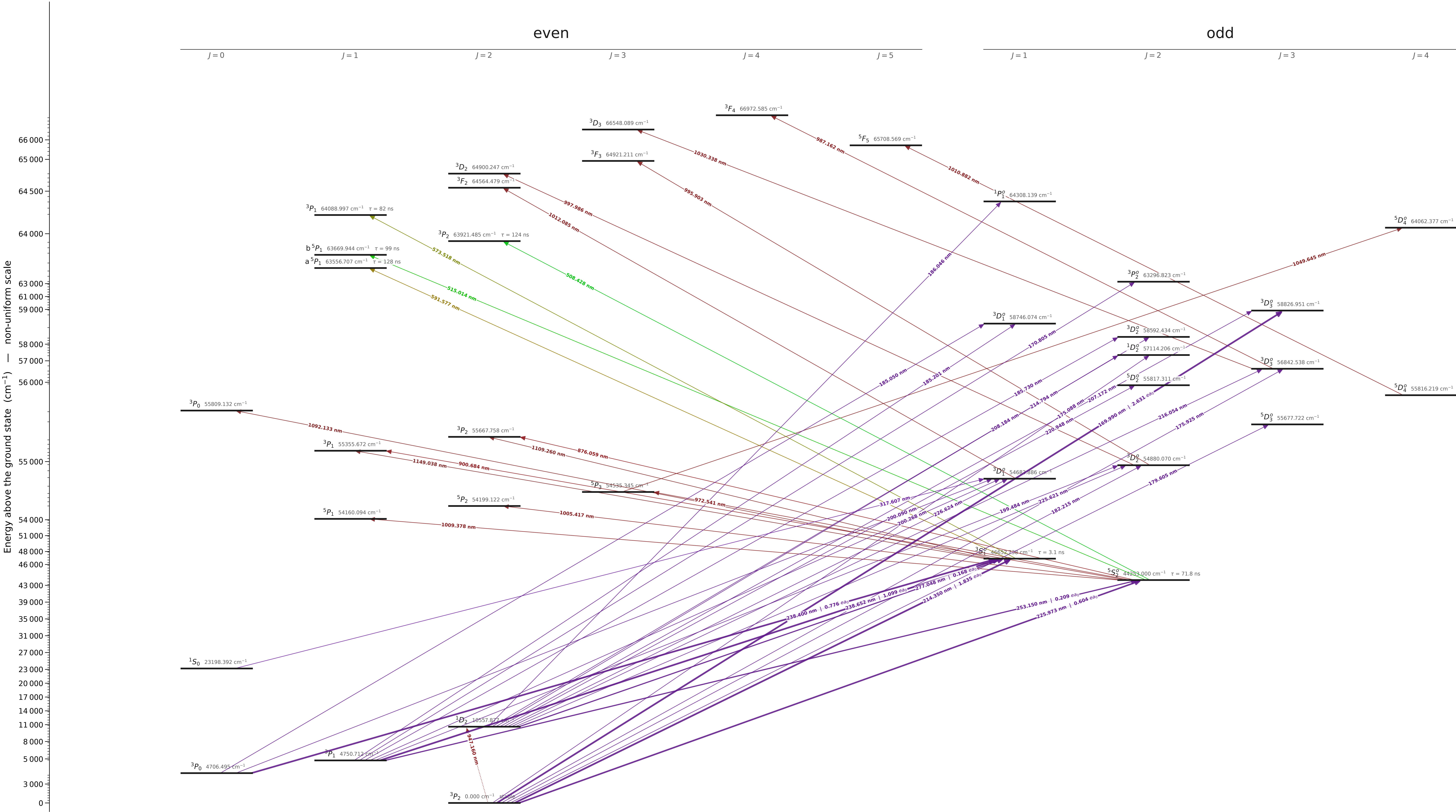
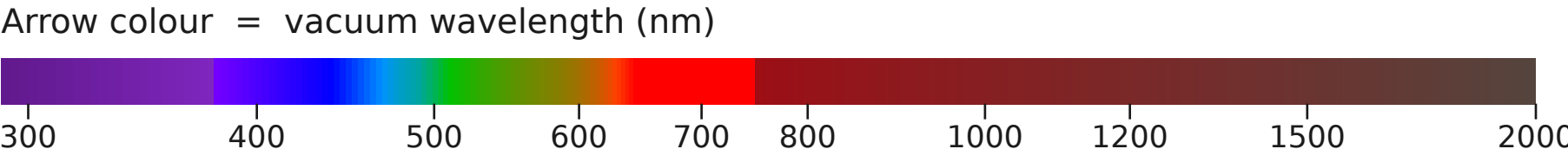


# Tellurium

Arrow label: vacuum wavelength | reduced dipole matrix element  $| \langle J || e r || J' \rangle |$  in  $e a_0$   
Matrix elements are measured or NIST values; \* = high-accuracy theory,  $\approx$  = model or semi-empirical calculation.  
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation,  $n = 70$ .  
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



## Key transitions

| Transition           | $\lambda$ vac (nm) | $\lambda$ air (nm) | $\nu$ (THz) | $  \langle J    e r    J' \rangle  $ ( $e a_0$ ) | A ( $s^{-1}$ ) | $\Gamma/2\pi$ up. (MHz) | branch (%) | Use                                                          |
|----------------------|--------------------|--------------------|-------------|--------------------------------------------------|----------------|-------------------------|------------|--------------------------------------------------------------|
| $^3P_2 - ^3S_2^o$    | 214.3497           | 214.2822           | 1398.6139   | 1.835                                            | 2.310e+08      | 51.34                   | 71.6       | first step of the ISOLDE RILIS resonance-ionization schem... |
| $^3S_2^o - ^3P_1$    | 573.5175           | 573.3585           | 522.7259    | -                                                | -              | 1.941                   | -          | second step of the ISOLDE RIUS resonance-ionization sche...  |
| $^3S_1^o - a\ ^5P_1$ | 591.5770           | 591.4132           | 506.7682    | -                                                | -              | 1.243                   | -          | second resonant step                                         |

$\lambda$ ,  $\nu$  from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.  
 $\Gamma/2\pi$ : natural linewidth of the upper level,  $1/(2\pi\tau)$ .

## <sup>125</sup>Te hyperfine structure ( $I = 1/2$ )

| Level     | A (MHz)    | B (MHz) | Shift of each F level from the centre of gravity (MHz) |
|-----------|------------|---------|--------------------------------------------------------|
| $^3P_2$   | -998(14)   | -       | F=3/2: +1497.00 F=5/2: -998.00                         |
| $^5S_2^o$ | 1010.4(13) | -       | F=3/2: -1515.60 F=5/2: +1010.40                        |
| $^3S_1^o$ | 943(20)    | -       | F=1/2: -943.00 F=3/2: +471.50                          |

Measured A, B constants; sources in the data file.

## Isotope shifts

| Transition        | $\lambda$ vac (nm) | Isotopes | Shift (MHz) |
|-------------------|--------------------|----------|-------------|
| $^3P_2 - ^3S_1^o$ | 214.350            | 120-130  | -1029(11)   |
| $^3P_2 - ^3S_1^o$ | 214.350            | 122-130  | -829(13)    |
| $^3P_2 - ^3S_1^o$ | 214.350            | 124-130  | -625(13)    |
| $^3P_2 - ^3S_1^o$ | 214.350            | 126-130  | -426(11)    |
| $^3P_2 - ^3S_1^o$ | 214.350            | 128-130  | -211(11)    |
| $^3P_2 - ^3S_1^o$ | 214.350            | 132-130  | 214(15)     |
| $^3P_2 - ^3S_1^o$ | 214.350            | 134-130  | 402(15)     |
| $^3P_2 - ^3S_1^o$ | 214.350            | 136-130  | 1242(21)    |
| $^3P_2 - ^3S_1^o$ | 214.350            | 123-130  | -815(15)    |
| $^3P_2 - ^3S_1^o$ | 214.350            | 125-130  | -594(19)    |

Shift =  $\nu$ (first isotope) -  $\nu$ (second isotope).